

MAX HORWITZ

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Education

University of Washington

3.91/4.00

B.S. Electrical Engineering & B.S. Mathematics

Rising Junior – Expected Graduation June 2028

Coursework (Graduate Level[†]): Advanced Probability[†], Optimization[†], Linear Programming, Statistical RL[†], Learning in Games[†], Machine Learning, Linear Systems[†], Control Theory[†], Algebraic Combinatorics[†]

Research

My research emphasises **designing algorithms with provable guarantees**. This involves using theory and experimentation in a closed loop - my background in mathematics gives me ideas for algorithms which I test in experiment, and the results of these experiments inform the guarantees I aim to prove.

Papers

(In Review) M. Horwitz, J. Gonzales, E. Mazumdar, L. Ratliff. “Structure from Strategic Interaction & Uncertainty: Risk-Sensitive Games for Robust Preference Learning.” *arXiv:2605.09946*, 2026

Using risk measures and game theory to design new algorithms for LLM fine tuning. Implemented a family of self play algorithms, studied the effects of debiasing methods, and evaluated on 8b+ parameter models. Contributed to sample complexity analysis in both the online and offline settings, incorporating bias and asymmetry introduced by risk measures.

(Accepted RLC) J. Gonzales, M. Horwitz, E. Mazumdar, L. Ratliff. “Strategically Robust Multi-Agent Reinforcement Learning with Linear Function Approximation.” *arXiv:2603.09208*, 2026

Proposing an alternative to the standard Nash solution in Multi-Agent RL. Implemented a novel optimistic value iteration algorithm, and used OvercookedV2 and a custom stag hunt environment to evaluate strategic robustness. Presented our work to an audience of 100+ at RLC

(Accepted Arnold Math. J.) Z. Chowdhury, F. Clément, M. Horwitz. “Obtaining the Chamanara Surface from the van der Corput Sequence.” *arXiv:2511.00311*, 2025

Explaining the behavior of graphs generated by certain low-discrepancy sequences. Contributed a proof of the torus shape of the Kronecker graph, and worked on the interval-exchange framework. Wrote visualization software. Presented our work at NUMS and UNC online undergrad research seminar.

Programs

SURF — Summer Undergraduate Research Fellowship

Caltech | June – Aug. 2026

Worked with Prof. Mazumdar, Prof. Wierman, and Prof. Murthy. Proposed and explored several ideas relating to control and steering of diffusion models. Designed algorithms, evaluated theoretically with tools from optimal control and optimization, and through experiments on image generation and robotics tasks.

WXML — Washington Experimental Mathematics Lab

UW | Autumn 2025

Worked with Dr. Clement and a group of other undergrads. Designed and wrote software modeling a variant of poker as a sequence of optimization problems. Read extensively up on optimization and game theory.

Other Experience

Extracurricular

EE510 — Teaching assistant for introductory graduate course on foundations of Systems Theory Autumn 2026

Competitive Programming — UW club member; top 0.1% on Project Euler 2024 – Present

EPS BCI — Team of 4 building EEG headset. I worked on circuitry, software, and signal processing 2022 – 2024

Skills

Programming: Python (ML packages, RL environments, Data analysis), Java, L^AT_EX

Tools: Slurm/HPC clusters, Git, Linux, conda, Huggingface